T243 | Wk 9 | S07 | GP-S404 | Sarah

P631: Accelerating opposition analysis in professional rugby

Past fortnight

Over the past fortnight, the Team focused on developing a clearer understanding of the project following our first meeting with the Industry Partner. We reviewed the Partner meeting information to document the current opposition-analysis workflow, identify key pain points, and determine what information is still required before we can confirm a technical direction.

As the sample CSV files and existing notebooks discussed during the Partner meeting have not yet been provided, the Team sent a follow-up email requesting an update. Rather than pausing work while waiting for the data, we identified activities that could be progressed using the information already available from the Partner meeting.

The Team organised the next stage of work into several areas, including the GitHub repository, Current Workflow Diagram, Technology Research and Justification, Pain Points and Open Questions, and Possible Prototype Directions. Responsibilities were distributed among Team members so that each person could contribute to the project and provide clear evidence of their individual work.

A shared GitHub repository was also established to organise project work and individual evidence. Initial work has begun on documenting the current workflow, identifying workflow problems and unanswered questions, and researching technologies that may support the project. These activities are intended to help us prepare for technical development once the Partner's data and existing code become available.

Fortnight ahead

Over the next fortnight, the Team plans to complete and consolidate the current project-discovery work, including the workflow documentation, pain-point analysis, technical research, open questions and possible prototype directions. Each Team member will continue uploading evidence of their contribution to the shared GitHub repository so that it can be referenced in our fortnightly documentation.

If the Partner provides the sample CSV files and existing notebooks, our priority will be to inspect the data structure, understand the current extraction process, and compare the actual implementation with our current understanding from the Partner meeting. We will

then use this information to validate the identified pain points and begin narrowing the possible technical and prototype directions.

If the files are still unavailable, the Team will continue progressing the work that does not depend on direct access to the data, including technical research, project planning, documenting assumptions and open questions, and considering which small working component could be suitable for the end-of-semester prototype demonstration.

We also plan to discuss the current dependency on the Partner's data with our Tutor and seek feedback on whether our current project direction and preparation are appropriate while we wait for access to the required files.

To discuss

- We have followed up with the Industry Partner regarding the sample CSV files and existing notebooks discussed during our first meeting, but we have not yet received them. We would like to discuss whether there are any additional tasks we should prioritise while waiting for access to the data.
- Based on the first Partner meeting, we have started documenting the current workflow, identifying pain points, researching possible technical approaches, and considering possible prototype directions. We would like feedback on whether this is the right level of progress for this stage of the project.
- We would also like to discuss how much detail is expected when identifying a possible working prototype for the end of Phase 1, given that we have not yet been able to inspect the Partner's actual data or existing code.
- We have started organising our work and evidence through the GitHub repository and individual contribution folders. We would like to confirm whether linking this evidence directly in the Week 9 documentation is sufficient for assessment purposes.

a. Kelvin Dang (n12051632)

My work

Over the past fortnight, I contributed to both the technical setup and early development of the project. I set up the team GitHub repository and created a clear folder structure so that each member can organise and contribute their work more easily. I also spent time researching rugby union and created a short PDF summarising the main rules to help the team better understand the sport and the context of opposition analysis. A significant part of my work was searching for suitable match data, especially detailed play-by-play and match event data that could be used for analysis. Since we have not yet received data from our partner, I had to look for external datasets so that I could start becoming familiar with the type of data we may work with later. I used this external data mainly to understand the data structure, explore possible analysis approaches and prepare the technical workflow in advance. Using this data, I developed a small working prototype that processes match events and produces an initial opposition analysis report, including basic event summaries and visualisations. I also tested and refined the workflow so that it can work directly with the available CSV datasets.

My plans

This work helped establish the technical foundation of the project and gave the team an early example of what an automated opposition analysis system could look like before we receive the official partner data. Over the next fortnight, I plan to continue preparing and improving the technical workflow while we are still waiting for the partner data. I will continue using external match data only as a temporary way to test ideas, improve the prototype and become more familiar with possible analysis methods. Once we receive the partner's data, I plan to move directly to analysing that dataset instead of continuing with external data. I will examine its structure, identify useful variables and patterns, and adapt the current prototype to better support the opposition analysis needs of the Queensland Reds. I also plan to discuss the results and possible analysis directions with the team and use feedback from our partner to decide which features and insights should be prioritised next.

b. Leon Duong (n11849789)

My work

Over the past two weeks, I mainly focused on researching technologies for our project. I looked at Python, Pandas, SQLite and Streamlit, as well as React and FastAPI as an alternative. I compared their benefits and limitations and collected some official sources that our team can use as evidence. I also looked more closely at the current workflow and how technology could help reduce repeated processing of old game data and push my works to our team GitHub repo.

My plans

Over the next two weeks, I plan to continue researching the technologies and see how they could work with our actual project data. I will discuss my findings with the team and help with choosing which technologies we should test. I also hope to do some basic testing to see which approach works better for our project.

c. Nao Hakoda (n11917580)

My work

Review the meeting note with partner:

I couldn’t join the partner meeting so I reviewed the meeting note based on the recordings.

Dimitri's workflow (fragmented multi notebook process, repeated processing of already processed games, and reading the same raw files multiple times) and Jack's workflow (time consuming Tableau setup and manual review of outputs)

Summarised why each one is a problem for them.

Pain Points/Problem List:

I reviewed the meeting note to identify the problems

Open Questions / Assumptions

Based on the meeting notes, I identified the information we are still missing because we haven't yet received the Partner's CSV files or notebooks. I grouped these into questions about the data structure, the existing notebook/code structure, and priorities/scope that still need clarification from the Partner.

My plans

in the next fortnight, I will plan to update these two sections once we receive the sample CSV files and notebooks from the Partner, since several of the open questions should be answered directly from that data.

d. Ngoc Ha Nguyen (n12248746)

My work

Over the past two weeks, I mainly focused on understanding the first Partner meeting and identifying possible directions for our prototype. I reviewed the current opposition preview process, especially how the team uses rugby data, CSV files and video analysis. I also looked at some of the current problems, such as repeatedly processing previous games and the manual work involved in preparing the analysis.

I also helped identify a few possible prototype ideas, including automating game extraction, creating an analysis pipeline and developing a simple dashboard. I organised the main points from the Partner meeting and prepared questions and follow-up information that could help our team better understand the project requirements.

My plans

Over the next two weeks, I plan to discuss the possible prototype directions with the team and decide which ideas are worth exploring further. I will also continue looking at the Partner's feedback and any additional information or data they provide.

I also plan to help document our communication with the Partner and refine the prototype ideas based on their requirements. If possible, I will start looking at how one of the simpler prototype ideas could be tested with the sample rugby data.

e. Star Kamonthitiwuth (n11864133)

My work

Over the past fortnight, I have mainly contributed to the team's documentation, coordination, and project planning following our first meeting with the Industry Partner.

After the Partner meeting, I used the meeting recording to prepare a written transcript so that the team could review the discussion accurately and refer back to the Partner's requirements, current workflow, pain points, and suggested next steps. I used this transcript as a reference when organising the information from the meeting and identifying what the team could continue working on while waiting for the sample CSV files and existing notebooks that the Partner had discussed providing.

As the Partner had not yet provided the additional data and files, I sent a follow-up email requesting an update. I also prepared two possible sets of next steps for the team: one for the situation where the Partner responded and provided the files by Tuesday, and another for the situation where we still had not received a response. This was intended to help the team continue making progress rather than stopping all work while waiting for the external dependency.

I then reviewed the advice from our Week 7 Tutor meeting and used it as a reference to organise the work that could provide useful evidence for our next fortnightly assessment. Sarah had emphasised the importance of showing evidence of project work, technical investigation, planning, and individual contribution, as well as beginning to consider what working part of the prototype could eventually be demonstrated.

Based on this, I prepared a task structure for the team consisting of five main areas: GitHub Repository, Current Workflow Diagram, Technology Research / Justification, Pain Points / Problems + Open Questions / Assumptions, and Possible Prototype Directions + Partner Communication Evidence. For each area, I added a short explanation of what should be included and how it related to the advice we received from our Tutor. I also suggested dividing the tasks between team members so that everyone would have a clear responsibility and evidence of their individual contribution.

I volunteered to take responsibility for the Current Workflow Diagram (evidence). Because we still do not have the Partner's actual code or sample data, I used the transcript from the first Industry Partner meeting to reconstruct the current workflow based on the information Dimitri and Jack provided. I developed separate current-state workflows for Dimitri's data-analysis process and Jack's performance and footage-analysis process, and identified the manual and repetitive steps within each workflow.

For Dimitri's workflow, I documented the process from game CSV files through data extraction, separate analysis notebooks, generated CSV outputs and figures, LaTeX preview generation, and the final PDF opposition report. I also identified issues such as repeated processing of previous games, repeated reading of the same raw files, running multiple notebooks separately, and manually switching between teams and notebooks. These issues were directly discussed during the Partner meeting.

For Jack's workflow, I mapped the statistics/data stream through Opta data, Tableau, statistics and visualisations, PowerPoint and key-stat review, together with the separate footage stream through SportsCode. I then showed how the statistics and footage are manually combined to produce the opposition preview for the coaches. I also documented pain points such as the time required to build Tableau outputs, manually review a large number of slides, identify key findings, and combine statistics with relevant footage.

Creating this diagram helped me develop a clearer understanding of the Partner's current process and where potential opportunities for automation may exist, even though we are still waiting for access to the actual project files.

My plans

Over the next fortnight, I plan to continue refining the Current Workflow Diagram and use it as a reference when the Partner provides the sample CSV files and existing notebooks. Once we receive these materials, I will compare the actual implementation against the workflow we reconstructed from the meeting and update any parts that are incomplete or inaccurate.

I also plan to help the team review the relationship between the current workflow, identified pain points, technical research, and possible prototype directions so that these areas remain consistent with each other. In particular, I want to make sure that any prototype direction we consider is clearly connected to a problem identified in the current process rather than selecting a solution before understanding the Partner's data and existing code.

If the Partner provides the files, I plan to support the team in documenting the CSV structure, current extraction process, and any additional workflow details that become clearer from the actual implementation. If we still do not receive the files, I plan to continue documenting the remaining assumptions, open questions, and dependencies so that we can clearly explain to our Tutor what information is still unavailable and what work the team has continued to complete in the meantime.

I have uploaded my Current Workflow Diagram and its supporting files to the Team's GitHub repository and linked the evidence directly in the Week 9 documentation. Over the next fortnight, I plan to continue maintaining and updating this evidence as the project develops. I will also continue helping the Team organise our next steps and narrow the possible prototype directions toward a small working component that could eventually be demonstrated at the end of Phase 1.